Ice Rumble

WORKING TITLE

A LAN party game for 2–4 players: short physics mini-games, one shared scoreboard.



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Concept render, AI-generated. Not gameplay.

How it plays: one room, short rounds, one scoreboard

2–4 friends in one room, each at their own PC. One PC hosts; the others join over the local network. **Everyone plays the same cube, in one of four colours.**

1 Lobby

Host starts a session and sets the playlist. Others join by local IP.

2 Round

The next mini-game loads. Some have rounds of their own.

3 Play

Host runs the physics. Clients send input and draw the result.

4 Place

Round ends. Placement becomes points.

5 Standings

Ends after a set number of mini-games or at a target score.

NEXT MINI-GAME

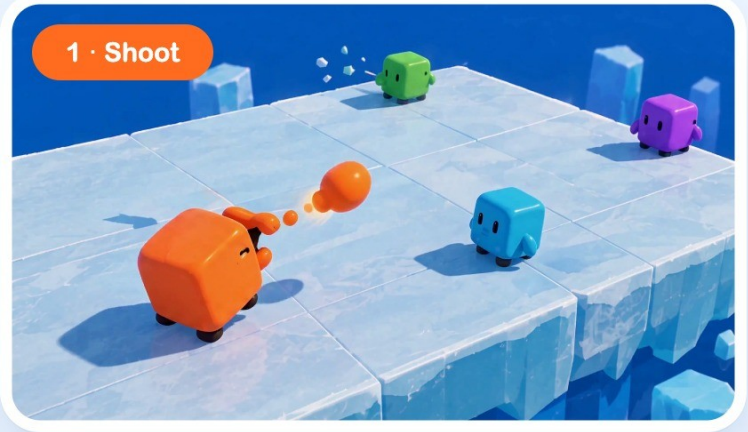
PLACEMENT POINTS · PROPOSED

1st	4	PTS
2nd	3	PTS
3rd	2	PTS
4th	1	PTS

BETWEEN ROUNDS

CARRIES OVER	Points. Nothing else accumulates.
PLAYLIST	Random or custom. Unwanted games can be left out; a game can be played more than once.
RESETS	Mass and position, so one bad round does not snowball.

Every hit makes you lighter, so the next one sends you further



Shots are real physical projectiles. Shooting is free; only a fire-rate limit sets the pace.

SHOOTER MASS

Unchanged: shooting costs nothing



The target is pushed back from the direction of the shot and loses mass. Its size stays the same.

TARGET MASS

Faded part: lost to this hit



Less mass means the same shot pushes further. A cube that falls off is out.

KNOCKBACK, SAME SHOT

Grows as mass drops

The force of a shot never changes. **The target's mass does**, so every hit makes the next one more dangerous.

Bars show direction only. Fire rate, mass loss and force are set in playtests.

One shot, a different effect in each mini-game

MOVE	CAMERA	SHOOT	JUMP
WASD	Third-person	Real projectiles, fire-rate limit	Only where the mini-game allows it



MVP

Knockback

Mass Knockout

The target is pushed back from the shot's direction and loses mass.



MECHANIC IDEA

Steer

Ice Race

Shoot forward while sliding and use the shot's momentum to change direction.



BACKLOG

Launch

Parkour Blast

The shot explodes and lifts you. Shoot the ground to jump.



BACKLOG

Damage

Versus

A hit lowers the target's health.

The shot is shared code. **Each mini-game decides what a hit does.**

From our design doc. Steering by shot momentum is a mechanic idea; values come from playtests.



FIG 2 · Orange fires; Cyan is hit near the edge. Concept render, AI-generated.

MINI-GAME 01

Mass Knockout

GOAL

Knock the other cubes off the flat platform.

SHOOTING

Free, with a fire-rate limit. Shots are real projectiles; the shooter is not affected.

HIT

The target is pushed back from the shot’s direction and loses mass, not size.

FALLING

A cube that falls off dies.

ARENA

One flat platform. Open edges, no walls.

MINI-GAME 02

Ice Race

GOAL

Pass the most checkpoints before the time runs out.

LOOP

The track loops. Every checkpoint you pass counts +1, lap after lap.

ICE

Slippery floor: carry speed through corners without sliding off.

RIVALS

Push others off the ice to gain an advantage.

FALLS

Fall into the water and restart at your last checkpoint.



FIG 3 · Racing a floating ice track; the HUD is part of the render. Concept render, AI-generated.

Mini-game backlog: three more ideas from our design doc



IDEA 01

Parkour Blast

SHOT: LAUNCH

- Cross a course by shooting and blasting off.
- Hold to charge a strong blast. You can’t move while firing, so you are exposed.
- No hiding spots. Falling off is out.
- If hitting is too easy, shots can arc under gravity.



IDEA 02

Musical Chairs

SHOT: PUSH

- Player-shaped holes open in the floor, each at a set angle. One fewer every round.
- Whoever is left out is crushed by a giant bulldozer.
- Shots push rivals, but a cube in a hole can’t be pushed out.



IDEA 03

Versus

SHOT: DAMAGE

- A straight head-to-head battle.
- Hits lower health.
- Still being designed.

Ideas, not in the MVP. Each one reuses the shared shot and plugs into the same mini-game contract.

One shared core; each mini-game plugs into it

MINI-GAMES

Mass Knockout
mass · knockback · fall-out

Ice Race
ice grip · loop track · checkpoints

Next mini-game
new rules + one arena

MINI-GAME CONTRACT

Setup(players) → Run() → Results() : placement

SHARED GAMEPLAY

Character | TPS camera | Shot | Rounds & playlist | Scoreboard | Lobby & UI

NETWORK

Host-authoritative LAN · Unity Netcode for GameObjects (candidate)

ENGINE

Unity · built-in rigidbody physics

WHY THIS SPLIT

Reuse

A new mini-game is new rules plus one arena. Character, camera, network and scores already exist.

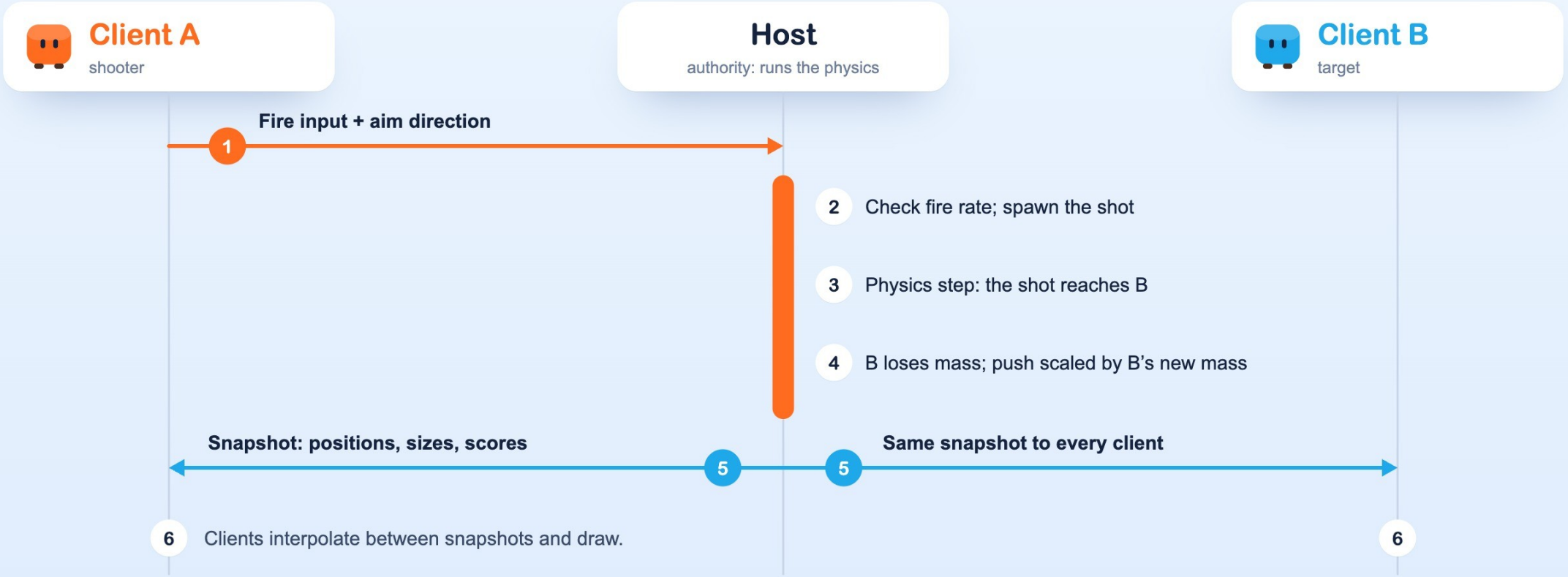
Per-game rules

Each mini-game sets what a shot does, how much grip the floor has, and whether jumping is on.

One truth

Everything that decides a result runs on the host, so every screen agrees.

One shot over the LAN, step by step



Your own movement is predicted locally, so it feels instant. Hits, mass changes and knockouts are decided only on the host. **Main risk:** keeping physics smooth on clients. LAN latency is low, and we tune the snapshot rate.

Social PC play is growing; LAN party games are rare

$43.6B

PC games revenue in 2025, up 12% year on year.

Newzoo [1]

$4.1B

Steam revenue of games with co-op in H1 2025, an all-time high (estimate).

Alinea [2]

70%

of US players aged 8+ play with other people.

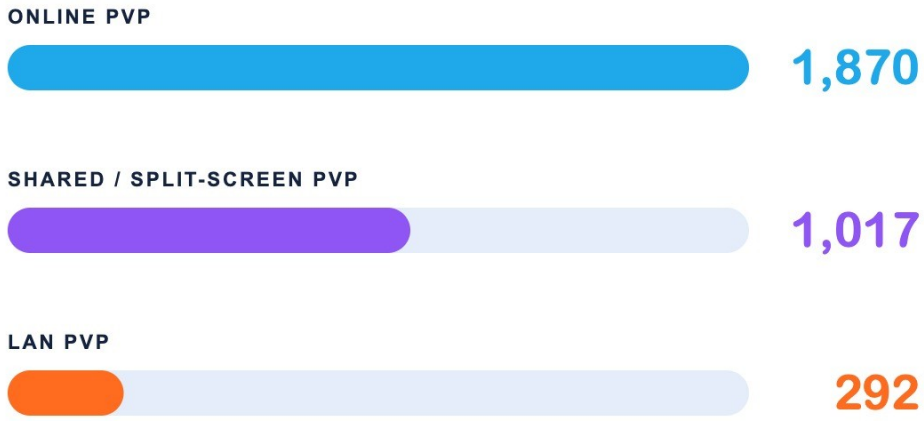
ESA 2026 [3]

≈8%

of ~3,490 Steam “Party Game” titles list LAN PvP.

Steam, Oct 2026 [4]

STEAM “PARTY GAME” TITLES BY MULTIPLAYER TAG



Steam store search, Oct 2026 [4]. Tags are user-applied.

RISK, STATED PLAINLY

An estimated 65.9% of 2025 Steam releases earned under $1,000 [5], while 5,863 games passed $100K in 2025 (Valve) [6]. **LAN is our differentiator, not a demand driver.**

Five games we are measured against



Party Animals

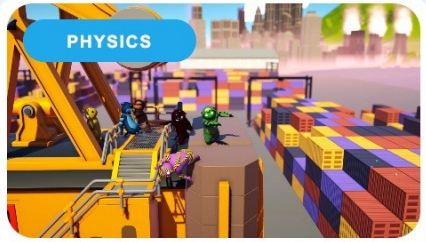
Recreate Games · 2023

VISUAL
Plush 3D animals, toy-like arenas.

MECHANIC
Physics brawling: grab, punch, throw rivals out.

MULTIPLAYER
Online and same-screen. No LAN listed.

FINANCIALS
Launch peak 104,174 players. Owner est. 2.5M–6.0M.



Gang Beasts

Boneloaf · 2017

VISUAL
Soft, gelatinous 3D people.

MECHANIC
Ragdoll brawling into stage hazards.

MULTIPLAYER
Same-screen and online. No LAN listed.

FINANCIALS
85% positive. Owner est. 4.6M–5.8M.



Fall Guys

Mediatonic / Epic · 2020

VISUAL
Candy-coloured beans, obstacle courses.

MECHANIC
Short race and survival rounds in a playlist.

MULTIPLAYER
Online, cross-platform only.

FINANCIALS
Free-to-play. Legacy Steam peak 172,213. Metacritic 80.



Pummel Party

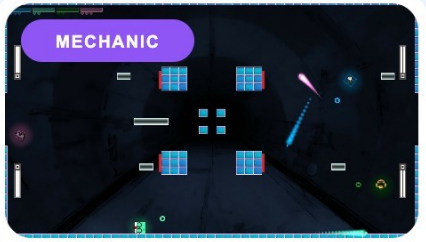
Rebuilt Games · 2018

VISUAL
Colourful low-poly 3D, slapstick items.

MECHANIC
Board game with short competitive mini-games.

MULTIPLAYER
4–8 online and local. No LAN listed.

FINANCIALS
$14.99. Peak 20,957. Owner est. 1.5M–7.6M.



Squarena

Codeaclysm · 2022

VISUAL
Minimal flat 2D squares.

MECHANIC
Arena shooter: low health shrinks you; small bodies feel more recoil.

MULTIPLAYER
Up to 6 local. No LAN listed.

FINANCIALS
$4.99. Sales not disclosed.

Screenshots: official Steam media, © publishers. “Owner est.” = third-party estimates via SteamDB, not sales. 2026-10-17 [7].

Where we sit among them

	PARTY ANIMALS	GANG BEASTS	FALL GUYS	PUMMEL PARTY	SQUARENA	ICE RUMBLE
3D characters	✓	✓	✓	✓	○	✓
Ranged attacks	–	○	○	–	✓	✓
Hits lower mass, so knockback grows	○	○	○	○	○	✓
Several mini-games, one result	○	○	–	✓	○	✓
Official LAN play	○	○	○	○	○	✓
Plays with no internet	○	✓	○	✓	✓	✓

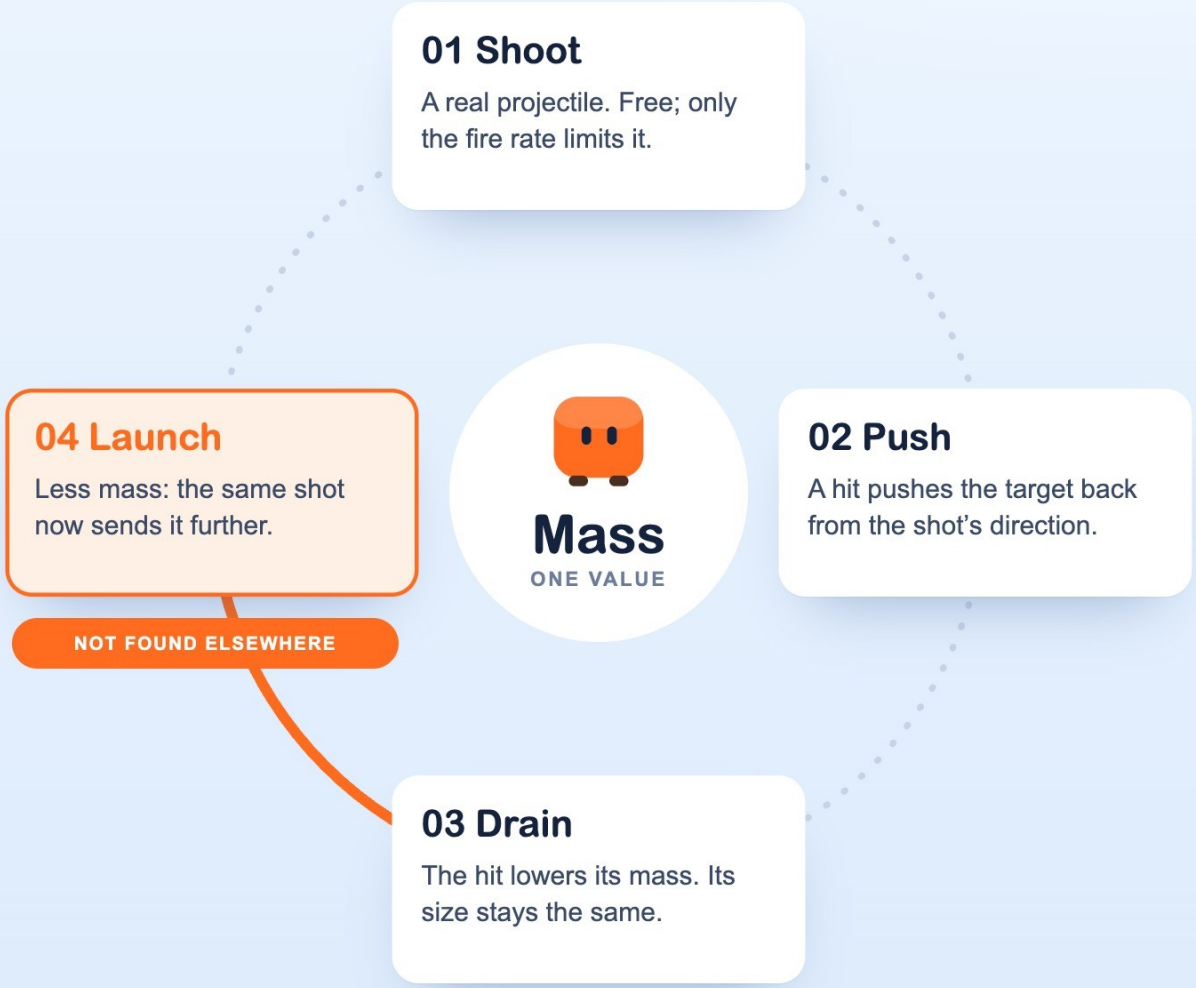
Yes Partly No Planned (Ice Rumble)

“Official LAN: No” = not advertised in the game's Steam features. “Partly”: Party Animals and Pummel Party have some weapons or items; Fall Guys' playlist eliminates rather than scores. Party Animals needs internet even on one screen [8]. Ice Rumble column = planned design.

THE NEW LINK

No shipped game we found ties knockback to the mass lost from hits.

Mass Knockout builds every hit on one physical value: mass. Links 01 to 03 have relatives in other games. **Link 04 we did not find.**



Prior-art search of shipped games, Oct 2026 [9–10]. Evidence on the next slide.

Closest prior art for each link, and what we did not find

LINK		CLOSEST PRIOR ART	STATUS
01–02	**Shots that push**	Knockback from attacks is standard in the genre, e.g. Smash Bros.	STANDARD
03	**Hits change your body**	Squarena (2D): lower health, smaller body. Morphies Law: hits move mass between robots.	PRECEDENT
04	**Lighter, launched further**	Smash Bros. raises knockback with a damage % counter, not mass. In Squarena, smaller bodies feel more of their own recoil.	NOT FOUND
01–04	**All four in one 3D party game**	No shipped game found.	NOT FOUND

Prior-art search of shipped games, Oct 2026 [9–10]. “Not found” means not found after searching.

Same rules, different builds: starting mass sets how far you fly



FIG 4 · Heavy, balanced and light builds of the same character. Concept render, AI-generated.

Weight classes are standard in the genre (e.g. Mario Kart). Here starting mass feeds the knockback rule directly, so a build changes how you play, not only how you look.

Heavy

More starting mass: harder to push off the edge.

STARTING MASS

KNOCKBACK TAKEN

Balanced

The reference build the others are tuned against.

STARTING MASS

KNOCKBACK TAKEN

Light

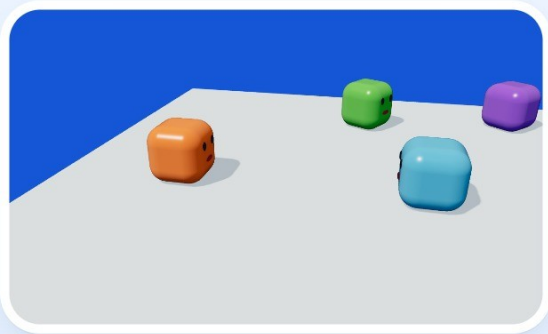
Less starting mass: flies further when hit, jumps higher where jumping is on.

STARTING MASS

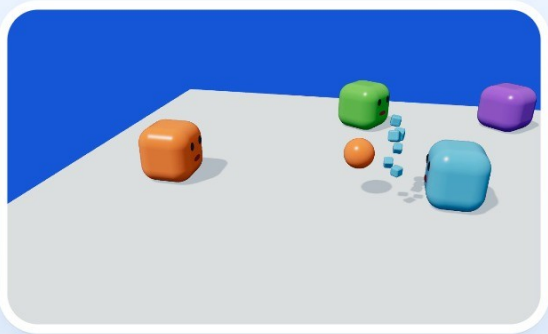
KNOCKBACK TAKEN

Idea from our design doc: selectable character types. Not in the MVP; pips are relative.

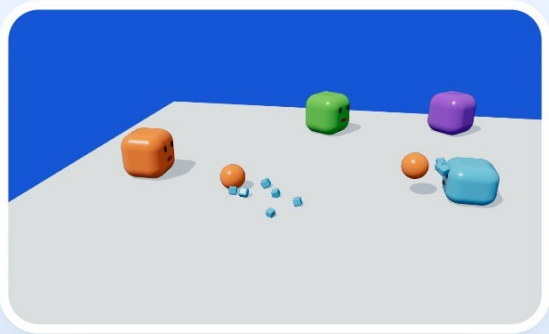
Prototype: we checked that the mass rule reads on screen



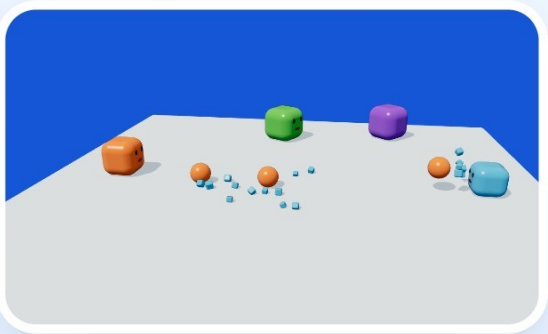
Before



Shot 1



Shot 2



Shot 3

WHAT WE BUILT

A deterministic physics simulation of the Mass Knockout rule in Three.js, rendered frame by frame.

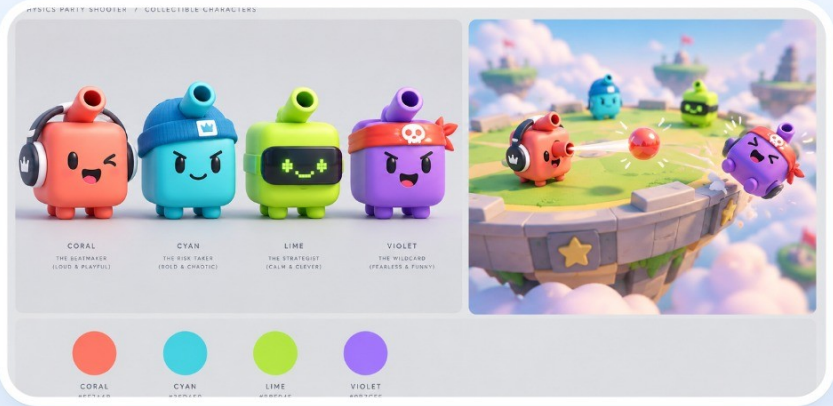
WHAT WE SAW

Over three shots the target flies further each time. It also shrinks here to make the rule visible; in the game only mass drops.

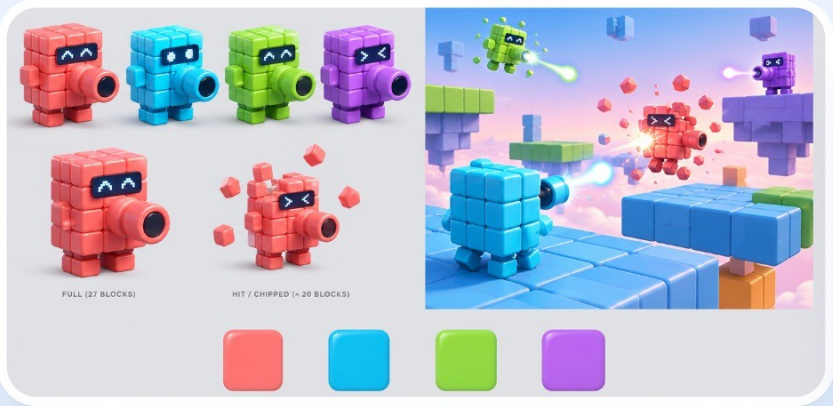
HOW WE USE IT

It is the reference for our first Unity test of Mass Knockout in Demo 1: same scene, same rule.

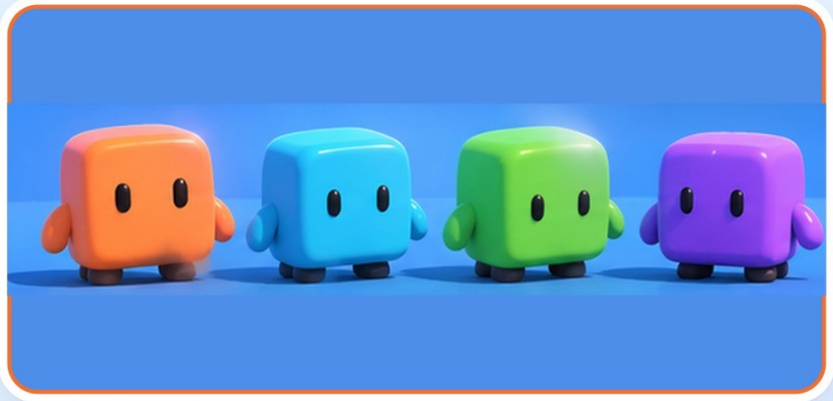
Four art directions for the same cube



A Vinyl toy



B Building blocks



C Glossy toy · chosen



D Jelly

PURPOSE

Four looks for the same character, generated to choose a direction before modelling.

USED IN THIS DECK

C, glossy toy. Every concept render in this deck uses it.

PLAYER COLOURS

Orange · Cyan · Lime · Purple

All AI-generated drafts. None is gameplay.

Logo drafts



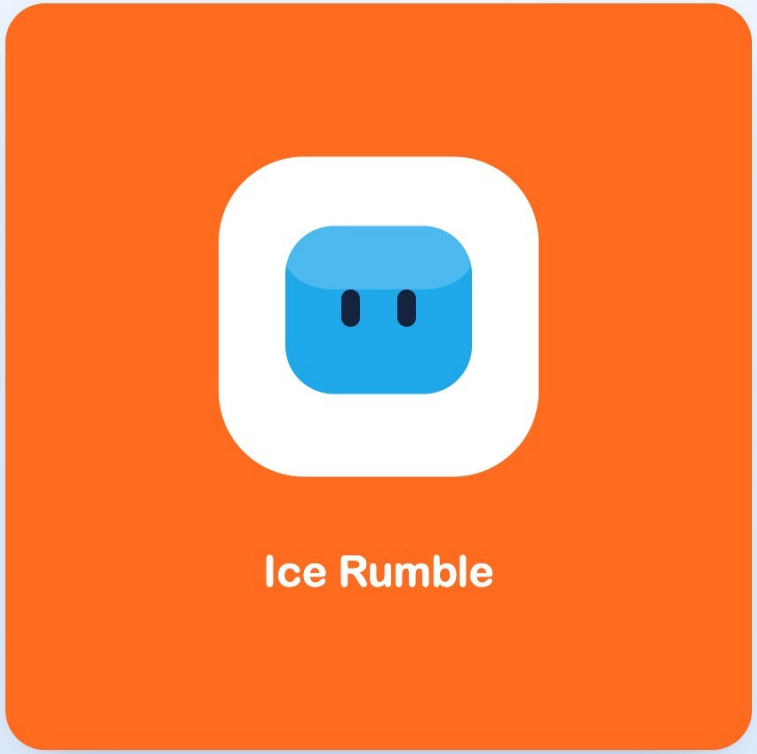
DRAFT A · WORDMARK

Title screen, deck cover, video end card.



DRAFT B · STACKED

Square formats: store capsule, lobby screen.



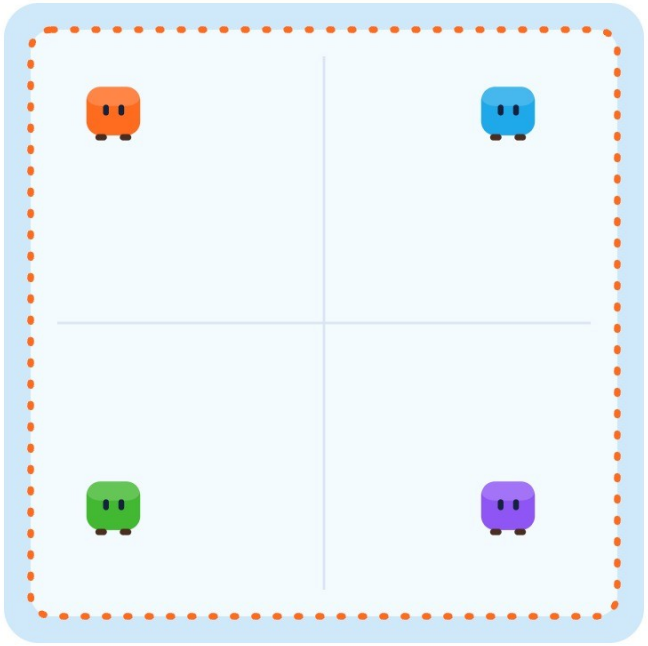
DRAFT C · ICON

Taskbar, build icon, favicon.

Drafts made for this proposal. The name is a working title; the logo is not final.

Level drafts, top-down: arena and track

MASS KNOCKOUT · ARENA DRAFT



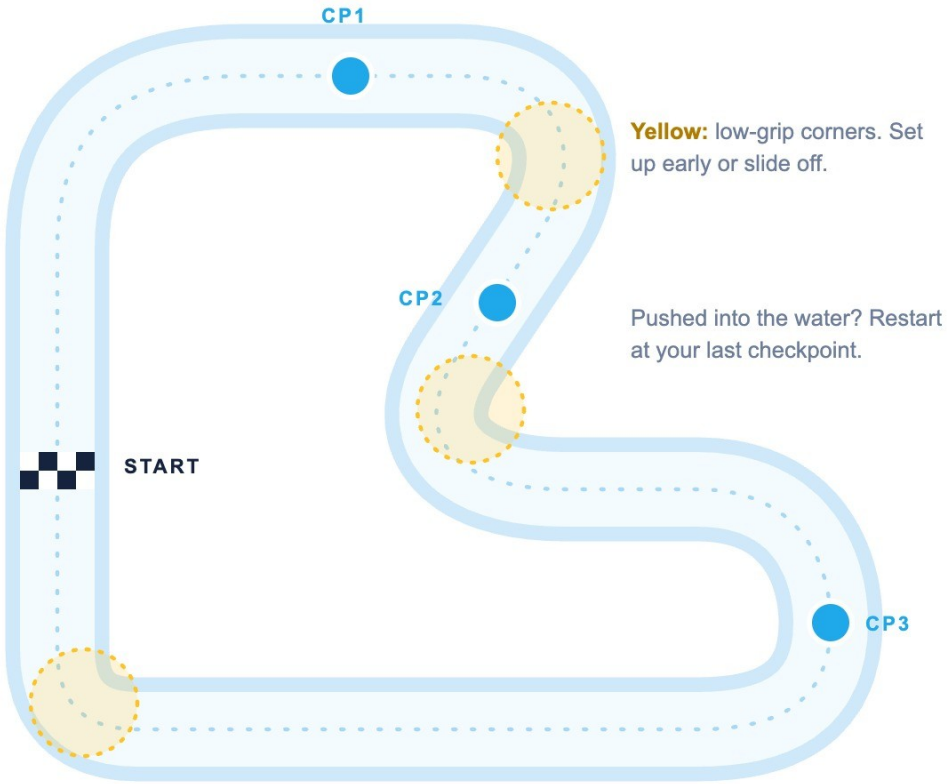
Fall zone. No walls, no cover.

One flat square platform.

Spawn near corners (draft).

OPEN EDGE: FALL OFF AND YOU ARE OUT

ICE RACE · TRACK DRAFT



Yellow: low-grip corners. Set up early or slide off.

Pushed into the water? Restart at your last checkpoint.

Draft layouts for blockout. The track loops: after CP3 you pass CP1 again, and every checkpoint counts +1. Sizes and checkpoint count are not set.

Roadmap: 12 weeks, four presentations, one playable step each

COURSE CHECKPOINTS · ALL MONDAYS

19 Oct

DONE

Proposal

Concept, research, prototype and drafts. This deck.

16 Nov

Demo 1

Shared core and LAN host/join. Ice Race playable as a blockout.

14 Dec

Demo 2

Mass Knockout over LAN. A full session: playlist, rounds and the scoreboard.

11 Jan

Final

Tuned, polished build. Promo video. Date subject to change.

RISK 01

Networked physics

Clients disagree on a knockback.

Host simulates; clients only render and interpolate.

RISK 02

Balance

Hits are too easy to land.

If playtests show it, shots arc under gravity, so aiming takes skill.

RISK 03

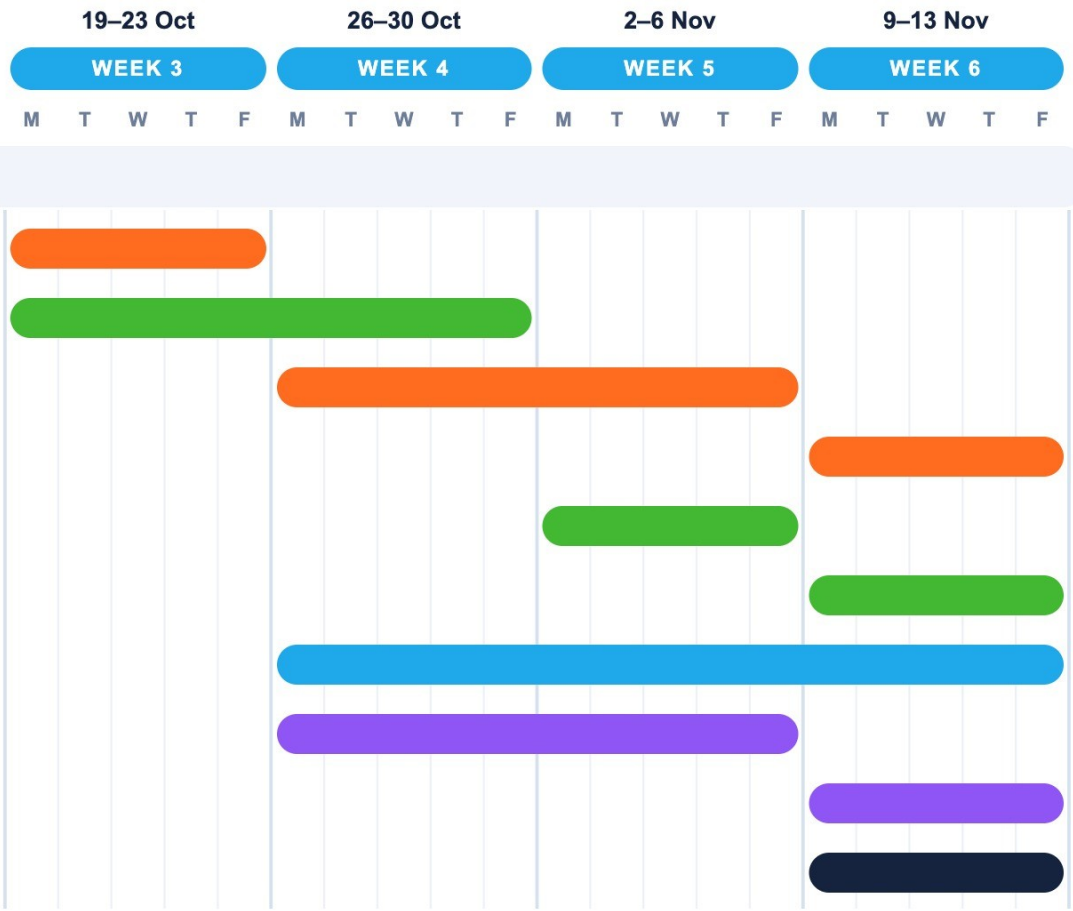
Scope

Too many mini-games for one term.

Exactly two in the MVP. More only once both are fun.

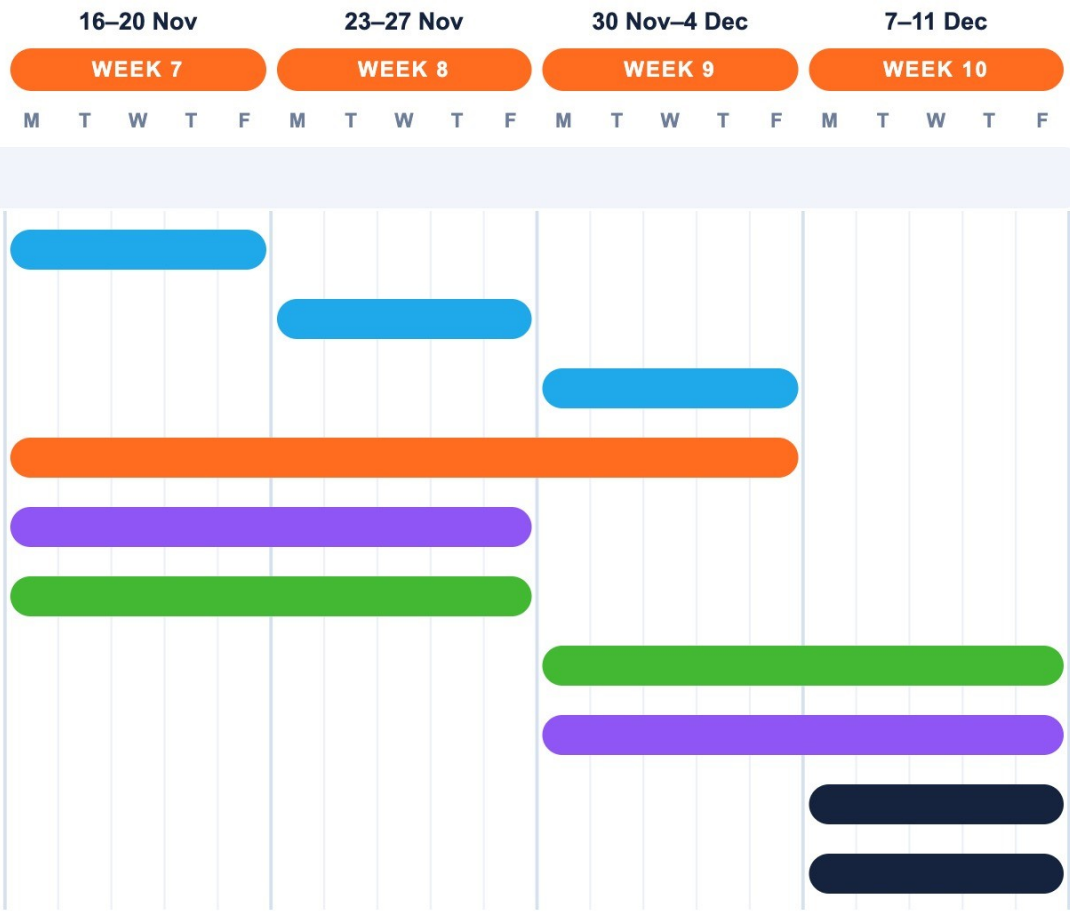
Gantt chart: demo 1, the shared core and Ice Race

WBS	TASK	OWNER	DURATION
2	**Demo 1 · 16 Nov**		
2.1	Unity project, scenes, build pipeline	Ezel	1 wk
2.2	Character controller & camera	Kürşat	2 wk
2.3	LAN host / join (Netcode)	Ezel	2 wk
2.4	Host authority & state sync	Ezel	1 wk
2.5	Ice movement profile	Kürşat	1 wk
2.6	Ice Race track blockout & checkpoints	Kürşat	1 wk
2.7	Shot projectile & fire-rate limit	Abdurrahim	3 wk
2.8	Lobby & join UI	Ardıl	2 wk
2.9	Character model, four colours	Ardıl	1 wk
2.10	Demo 1 build & presentation	Team	1 wk



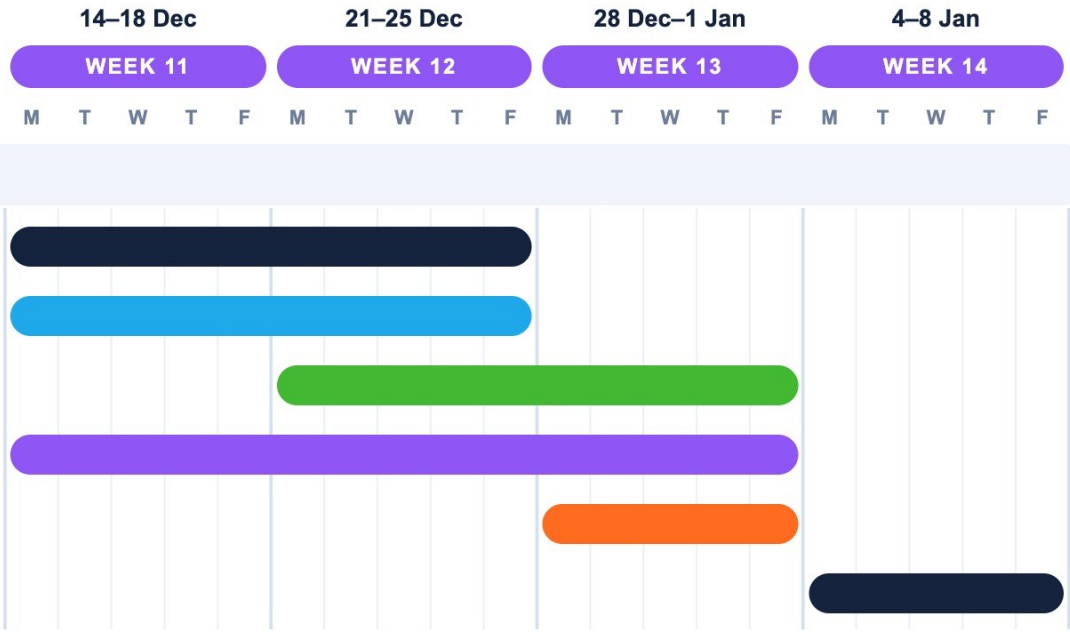
Gantt chart: demo 2, Mass Knockout and the full session

WBS	TASK	OWNER	DURATION
3	**Demo 2 · 14 Dec**		
3.1	Hit detection & mass loss	Abdurrahim	1 wk
3.2	Knockback scaled by mass	Abdurrahim	1 wk
3.3	Fall-out & placement order	Abdurrahim	1 wk
3.4	Networked projectiles & hits	Ezel	3 wk
3.5	Mass Knockout arena	Ardıl	2 wk
3.6	Ice Race loop & checkpoint scoring	Kürşat	2 wk
3.7	Playlist, rounds & end condition	Kürşat	2 wk
3.8	Scoreboard & results	Ardıl	2 wk
3.9	First full-session playtest	Team	1 wk
3.10	Demo 2 build & presentation	Team	1 wk



Gantt chart: final, tuning, polish and the promo video

WBS	TASK	OWNER	DURATION
4	**Final · 11 Jan**		
4.1	Balance tuning from playtests	Team	2 wk
4.2	VFX: trails, hit flash, splashes	Abdurrahim	2 wk
4.3	Sound effects & music	Kürşat	2 wk
4.4	Art & UI polish	Ardıl	3 wk
4.5	LAN stress test & final build	Ezel	1 wk
4.6	Promo video & final deck	Team	1 wk



PROPOSED OWNERS

- **Ezel** — Network, build
- **Abdurrahim** — Mass Knockout
- **Kürşat** — Movement, Ice Race
- **Ardıl** — Art, UI, session
- **Team** — Shared tasks

FINAL STANDINGS

Ice Rumble

1st 2–4 players, one LAN, no internet.

2nd Two mini-games on one shared Unity core.

3rd Every hit makes the target lighter, so the next one sends it further.



Concept render, AI-generated. Not gameplay.

Questions?

Ezel Bayraktar · Abdurrahim Özağaç · Kürşat Hürkan Aydın · Ardıl Bilal Ülkü

Sources

[1] Newzoo final 2025 figures, via GameDev Reports, 2026-06-26. gamedevreports.substack.com/p/newzoo-gaming-market-surpassed-200

[2] Alinea Analytics, games with co-op on Steam, H1 2025 (proprietary estimates). alineaanalytics.substack.com/p/games-with-co-op-generated-over-4

[3] ESA Essential Facts 2026 (survey Feb 2026, n=13,545). theesa.com

[4] Steam store search, tag “Party Game” × LAN PvP / Online PvP / Shared-Split-screen PvP, 2026-10-17. Tags are user-applied.

[5] Game Developers, 2025 Steam revenue distribution (Gamalytic estimates), 2026-08-21. game-developers.org/2025-steam-game-revenue-distribution

[6] PC Gamer, Valve at GDC 2026: 5,863 games over $100K on Steam in 2025. pcgamer.com

[7] Steam store pages and SteamDB charts, apps 1260320, 285900, 1097150, 880940, 1858400: prices, peaks, reviews, owner estimates. 2026-10-17.

[8] GameRevolution / Prima Games: Party Animals requires an internet connection, including split-screen.

[9] Squarena on Steam (app 1858400); codeaclysmgames.com/squarena-details.

[10] Morphies Law official site, morphieslaw.com/game.

Images: all game visuals are AI concept renders (GPT Image via Higgsfield; MiniMax H3) from our own reference art, except competitor screenshots (official Steam media). Logo and level drafts made for this proposal.